SA-CMF: A Secondary Attention-Based Cross-Modal Fusion Framework for Deep-Space Target Recognition in Remote Sensing

Na Li, *Member, IEEE*, Shuoyu Zhang, Huijie Zhao, *Member, IEEE*, and Wen Ou

***Abstract*—Deep-space infrared target recognition is significantly challenged by similar interference sources and complex backgrounds, which degrade detection accuracy (Acc) and robustness. While detection has shifted from single-detector to multidetector coordinated systems, current algorithms fail to consider both the dependencies between different features and the complementarity of the unique characteristics inherent to each modality, limiting fusion quality. To address this, we propose a secondary attention-based cross-modal fusion (SA-CMF) framework. The first attention stage applies modality-specific attention to enhance the discriminative ability of individual features, fully exploiting each modality's unique characteristics, while the second stage employs cross-attention to capture interdependencies among modal features. Additionally, a dynamic mutual information (MI) adjustment strategy is introduced to suppress redundant information and mitigate cross-modal distribution discrepancies. The experimental results demonstrate that SA-CMF achieves 94.7% Acc, high *F*1 scores, and robust performance under diverse deep-space conditions. These results validate the framework's effectiveness in improving both feature quality and target recognition, highlighting its potential for complex infrared remote sensing applications.**

***Index Terms*—Cross-modal fusion, deep space, infrared dim target, mutual information (MI) adjustment, remote sensing.**

I. Introduction

FROM the 1960s to 1980s, deep-space exploration and target detection systems were in their early stages, relying mainly on single platforms. Limited technological development hindered mid-course detection, making it difficult to meet application needs in complex backgrounds. Over time, single platforms evolved into multiple collaborative systems to acquire target characteristics and address higher application demands. Space-based space surveillance (SBSS) systems [1] enhance detection by acquiring temperature fields, radiation intensity, and infrared spectra. The space surveillance network (SSN) [2] captures multimodal features such as radiation intensity and reflectivity, while the space tracking and surveillance system (STSS) [3] enables continuous mid-course tracking with multiple infrared sensors. The Missile Defense Agency (MDA) [4] integrates systems like Terminal High Altitude Area Defense (THAAD) System for accurate detection based on diverse target characteristics. Given the heterogeneity of feature distribution and representation, leveraging multimodal information has become a key research focus in deep-space surveillance. However, multimodal feature fusion often struggles to simultaneously exploit modality-specific characteristics and capture intermodal correlations, leading to limited effectiveness of the fused representations.

In the context of this study, in the multimodal data obtained from multiple detectors on the sensing platform, temperature sequence features and spectral features are 1-D features, while radiative intensity features and motion features are 2-D features. Simultaneously, the pronounced heterogeneity in feature dimensionality and statistical distribution limits the effectiveness of conventional multimodal fusion strategies, which often fail to jointly preserve modality-specific discriminative cues and capture their coupled interactions.

Cross-modal fusion methods aim to achieve comprehensive integration [5], [6], [7], [8] of multimodal features by exploiting the mutual guidance and interaction between different modalities during the fusion process [9], [10], [11], thereby facilitating better utilization of multiplatform target characteristics [12], [13], [14], [15]. Various feature alignment strategies have been proposed to reconcile distribution and representation mismatches among modalities. However, feature fusion remains prone to redundancy, which hinders recognition accuracy (Acc) and increases computational cost [16], [17], [18], [19]. To mitigate feature redundancy, various strategies such as attention mechanisms, global–local feature integration, and structured feature matrices have been adopted.

As shown in Fig. 1, spectral feature (emissivity), as an intrinsic material-dependent property, provides a reliable basis for distinguishing different targets, while temperature acts as a key physical indicator in the deep-space environment. Radiation intensity features further capture target-specific variations, and motion features help separate targets from the background. Although these four feature types follow different

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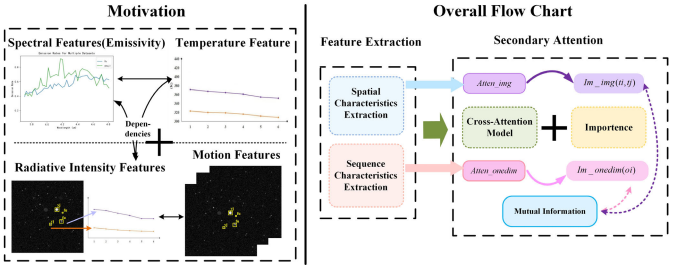


Fig. 1. Motivation and overall flowchart of the secondary attention-based cross-modal.

distributions, each makes a distinct contribution to recognition. Moreover, spectral feature (emissivity) and temperature are inherently coupled with radiation intensity, indicating inter-feature correlations. Consequently, given their heterogeneous distributions and representations, and considering that spectral feature (emissivity) and temperature are correlated with radiation intensity, it is crucial for the model to not only exploit the distinctive characteristics of each feature but also fully explore their coupling relationships, which poses a significant challenge for existing fusion strategies.

However, most existing multimodal fusion methods either treat different modalities with uniform importance or rely on fixed fusion strategies, which makes it difficult to simultaneously preserve modality-specific discriminative cues and effectively model their coupled relationships. As a result, informative yet secondary modal characteristics are often suppressed, and redundant correlations may be excessively amplified, limiting the robustness and effectiveness of the fused representation in complex deep-space scenarios.

Consequently, we propose a secondary attention-based cross-modal fusion (SA-CMF) framework for effective integration of multiplatform features. Attention weights of 1-D and 2-D features are first obtained, and then secondarily fused with cross-attention importance. This design fully exploits modality-specific properties while capturing intermodal dependencies and decoupling their mutual influences. Moreover, dynamic adjustment of mutual information (MI) mitigates redundancy and deviations caused by heterogeneous feature forms and lengths, yielding more effective spatial target representations. We summarize our contributions as follows.

1) We highlight the crucial role of modality-unique characteristics in cross-modal feature fusion. Each modality is first examined independently to capture its distinctive properties and to evaluate its relative contribution, thereby enabling a more effective exploitation of complementary information across modalities.
2) We design a secondary attention mechanism that integrates modality-specific importance with a cross-attention structure. The cross-attention captures intermodal dependencies while decoupling mutual influences across modalities. Combined with modality-unique characteristics, this mechanism enhances the discriminative power of the fused representation.
3) We introduce MI into the fusion process to model the interdependence between modalities. This enables dynamic adjustment of feature importance during fusion, allowing the model to better suppress redundancy and enhance informative representation.

The rest of this article is organized as follows. Section II reviews related work on cross-modal feature fusion. Section III details the proposed SA-CMF model. Section IV reports the experimental results, and Section V concludes this article. The code of the proposed SA-CMF framework is publicly available at: https://github.com/zhangshuoyu0729/SA-CMF-Cross-Modal-Fusion-Framework.git

II. Related Works

To better motivate our proposed method, this section presents a review of the current state of research in the field, focusing on three main aspects: the prevailing trends and challenges in cross-modal fusion and multiattention algorithms, feature acquisition and dim target recognition methods, and the existing issues associated with MI.

A. Cross-Model

Cross-modal fusion algorithms are generally categorized into three types. Preliminary fusion integrates multiple data sources into a single feature vector for neural networks, enabling mutual guidance between modalities. Late fusion trains modality-specific models and then combines their outputs in a fusion module. Intermediate fusion performs integration in intermediate network layers, often using a shared representation layer to merge features.

Early fusion combines features at the feature level via mapping, attention [20], [21], or spatial alignment [22], [23], using methods such as multiattention memory networks [24], spatial-size self-attention [25], and Transformer-based contextual attention [26]. It may introduce redundancy and fail to capture complementarity. Late fusion merges outputs from modality-specific classifiers [27], [28], but ignores interactions during feature extraction and depends on classifier performance. Intermediate fusion leverages joint interactions and attention mechanisms, including cross-modal skip connections [29], multimodal fusion with guided pooling [30], and hierarchical feature interaction [31], [32].

Effective feature alignment is essential for multimodal fusion due to varying feature distributions across modalities. Previous works have used multilayer Transformers and cross-modal attention [33], [34], or contrastive pretraining and semantic encoding [35]. However, these approaches still suffer from feature redundancy. More recent studies focus on cross-modal complementarity, including shared channel attention and adaptive fusion [36], dual cross-attention Transformers [37], and channel enhancement methods [38], [39], [40]. For instance, Liu et al. [41] adopted a spatial–spectral hybrid framework by independently extracting modality-specific features and fusing them via convolution, while Ding et al. [42] modeled superpixels as graph nodes to convert 2-D spatial information into graph representations and enhanced feature quality through contrastive multiview fusion. Nonetheless, integrating features effectively while suppressing redundancy remains a key challenge.

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Nevertheless, despite the critical role of feature coordination in cross-modal fusion, two key challenges remain: 1) existing methods for modeling intermodal correlations are underdeveloped, lacking systematic frameworks to jointly address intrinsic characteristics and mutual dependencies, which limits interpretability and performance and 2) these methods also struggle to achieve effective fusion while avoiding redundancy-induced interference.

These limitations motivate the need for a fusion framework that can explicitly preserve modality-specific characteristics while adaptively modeling intermodal dependencies without introducing excessive redundancy.

B. Multiattention Mechanism

Methods that combine channel- and spatial-attention mechanisms [43], [44] have been employed to extract more discriminative target features. Building on this, subsequent approaches [45] introduced pixel-level attention to further enrich feature representations, while others [46] enhanced semantic associations by integrating global and local attention mechanisms. Azeem et al. [47] develop a prototype-guided multilayer alignment network to improve few-shot object detection in remote sensing images. Multiattention has indeed shown strong capability in capturing diverse forms of features. However, its potential has not yet been fully explored in multifeature cross-modal fusion.

Meanwhile, multistage attention has been applied to capture multiscale target features. For example, Zhao et al. [48] enhanced multiscale feature learning by using fine-grained region localization to guide coarse-grained regions, while Yang et al. [49] constructed a secondary attention mechanism to emphasize valuable features and suppress noise. Other approaches [50], [51] optimize model representations by capturing diverse feature forms. Li et al. [52] propose a lightweight multiscale and multiattention network with multistage search for hyperspectral image classification, and Shu et al. [53] introduce multi-stage multi-attention (MSMA) for robust multimodal skin lesion classification. Nevertheless, current multistage attention methods still struggle to fully exploit modality-specific characteristics and interfeature interactions, which are critical for attention weight distribution.

Unlike prior hierarchical attention methods, our framework simultaneously captures intermodal dependencies, decouples their mutual influences, and preserves modality-specific characteristics through cross-attention, secondary attention, and adaptive weighting, thereby offering unique advantages for multimodal fusion in deep-space scenarios.

C. Feature Acquisition

In multimodal fusion, the choice of feature extraction strategies for each modality is crucial for effective and reliable cross-modal representations. For visual modalities, Transformer-based methods such as Swin Transformer and vision transformer (ViT) capture long-range spatial dependencies and model hierarchical local structures, respectively [54], [55], while ConvNeXt enhances convolutional architectures with Transformer-inspired training to yield stable and efficient 2-D feature representations [56]. For textual modalities, large language models (LLMs) provide discriminative 1-D embeddings that capture high-level semantics [57], [58], whereas traditional sequences feature extraction methods can provide complementary 1-D form-based features [59].

Despite these advances, existing approaches still face challenges in representing small spatial targets. These methods may lack sufficient local sensitivity, leading to suboptimal cross-modal correspondence for fine-grained spatial details.

D. Mutual Information

In cross-modal fusion, reducing redundancy is crucial to maximize useful information while minimizing unnecessary computational and representational overhead. Recent methods address this in different ways. CSFMamba [60] uses a cross-state fusion operator to selectively integrate multimodal features and suppress redundancy. RSRWKV [61] introduces a linear 2-D attention mechanism that focuses on essential spatial dependencies. Tensor-product-based multimodal networks [62] capture structured interactions between modalities, efficiently mitigating redundant feature contributions.

MI quantifies the correlation between different modalities, and fully leveraging this correlation is crucial in the modal fusion process. Additionally, MI aids in feature selection and effectively prevents feature redundancy during cross-modal fusion in multimodal data. Han et al. [63] proposed an MI structure that uses hierarchical operations to maximize the MI between single-modal input pairs (intermodal) and between multimodal fusion results and single-modal inputs, thereby preserving task-related information through multimodal fusion; Zhou et al. [64] applied MI minimization to explicitly encourage complementary information learning, thereby reducing redundancy and improving model performance; Zhang et al. [65] leveraged MI between model-based cross-modal inputs and outputs to derive unsupervised optimization objectives, and incorporated matching information based on MI into prototype contrastive learning to minimize both intramodal and cross-modal entropy.

However, even with the addition of MI to guide modality in the cross-modal fusion process, the unique characteristics of individual modalities are not incorporated into the fusion.

E. Infrared Dim Target Recognition

In deep-space environments, infrared point targets are typically dim and small, which significantly challenges detection and recognition. Existing algorithms include convolutional neural networks (CNNs), temporal neural networks, graph neural networks (GNNs), and Transformer architectures. CNN-based methods [66], [67], [68] extract local spatial features from targets and backgrounds, but rely on single-modality features, which is often insufficient for dim targets due to weak infrared characteristics. Temporal neural networks [69], [70], [71], [72] leverage sequential information to improve robustness; however, they struggle with similar co-moving objects, where trajectory similarity impedes discrimination. GNN-based approaches [73], [74] model targets as graph structures with internode relationships, yet these

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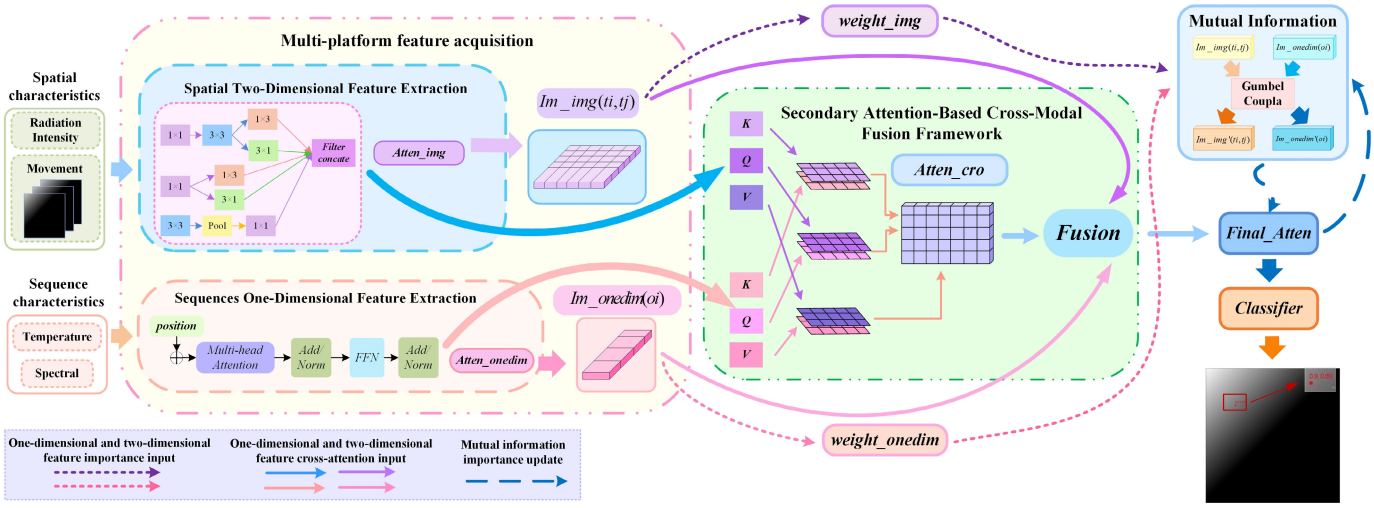


Fig. 2. Multimodal feature cross-modal fusion recognition flowchart. (The yellow regions represent the modality-specific attention modules and importance estimation for each modality. The green area denotes the cross-modal attention module. Together, these two components form the secondary attention mechanism. The blue region indicates the MI module, which assists in reducing the redundancy of the fused features.)

relationships are usually abstract and independent, failing to capture intrinsic correlations present in multimodal features. Transformer-based methods [75], [76], [77] use self-attention to model long-range dependencies, addressing convolutional locality limitations.

While some explore multimodal integration, the underlying physical mechanisms and intermodal correlations remain underexplored. To address these issues, this article proposes a dual-attention mechanism that emphasizes unique characteristics of each modality and exploits latent intermodal relationships, enabling more accurate recognition of spaceborne targets with enhanced feature representations.

III. Methodology

The model first separately extracts the sequence 1-D features and the spatial 2-D features, and then calculates the initial weights through the sequence and spatial feature attention mechanisms to obtain their respective weight distributions. Then, the cross-attention mechanism is applied to combine the importance of each feature, achieving a more accurate alignment of the sequence and spatial features. Meanwhile, MI is introduced to adjust in real time, reducing the impact of feature redundancy during fusion and effectively supplementing the spatial target feature information. The flowchart is shown in Fig. 2.

For clarity, Algorithm 1 provides a pseudocode description of the overall workflow and key components of the proposed SA-CMF framework.

Algorithm 1 Secondary Attention Fusion With MI Dynamic Intermodal Regulation

Input: spatial feature x, sequence feature y'
Output: final attention representation $Final_Atten$
Modality-specific attention modeling
Compute independent attentions:

$$Atten_img(x), Atten_onedim(y')$$

Estimate modality importance weights Im_img, Im_onedim.
Secondary attention fusion
Compute cross-modal attention $Atten_cro$.
Fuse modality-specific and cross-modal attentions $Fusion_Atten$.
Dynamic regulation via inter-modal dependency
Estimate inter-modal dependency according to the MI-based formulation in the main text.
Regulate secondary attention to obtain the final attention:

$$\begin{aligned} Final_Atten = {} & \alpha \cdot \left(Atten_cro\,(x) \oplus Atten_cro\,\left(y'\right)\right) \\ & + \beta \cdot \left(\frac{\exp\left(Im_img'\left(t_i, t_j\right)\right)}{\sum \exp\left(Im_img'\left(t_i, t_j\right)\right)} \cdot V_x \right. \\ & \left. \oplus \frac{\exp\left(Im_onedim'\left(o_i\right)\right)}{\sum \exp\left(Im_onedim'\left(o_i\right)\right)} \cdot V_{y'} \right). \end{aligned}$$

Return $Final_Atten$

A. Spatial 2-D Feature Extraction

Both the radiative intensity features and motion features exhibit spatial 2-D characteristics. Their characteristics distribution and variations are of significant importance for the study of space target properties.

Considering the characteristics of space target spatial 2-D data, InceptionV3 is used for feature extraction. When dealing with sequential 2-D observations, the same spatial feature extraction strategy is applied to each frame, allowing frame-wise spatial variations induced by target motion and radiative changes to be effectively captured. The key feature of the inception network is the expansion of convolution operations between layers. By factorizing the 7×7 convolution into two 1×7 and 7×1 convolutions, it increases the network depth, accelerates computation, and enhances nonlinearity. In InceptionV3, the 5×5 convolution kernel is replaced by two

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3×3 convolution kernels. Since large convolution kernels can be replaced by multiple 3×3 kernels, an $n \times 1$ convolution kernel is used in place of the 3×3 kernel.

The spatial 2-D features extracted by InceptionV3 are denoted as F_img_t, which are then processed by a soft attention mechanism to obtain the final feature representation

$$F_\text{img}_t = \text{Atten}_\text{img}(x). \tag{1}$$

After the F_img_t is obtained, a soft attention mechanism is applied. Unlike hard attention, which only adds attention to a few values, the soft attention mechanism allocates attention according to importance. The hard attention mechanism is not suitable for small target scenarios and will cause attention drift

$$\text{Atten}_\text{img}(x) = \sum_{i=1}^{K} \left(\frac{\exp(\text{similar}_i)}{\sum_{j=1}^{K} \exp\left(\text{similar}_j\right)} \cdot \text{Value}_i \right). \tag{2}$$

Among them, when the spatial 2-D feature is the radiation intensity feature, Value_i is the simulated radiation intensity of each group of spatial target characteristics corresponding to each frame, which serves as the reference value. Attention is allocated by comparing the similarity between the reference radiation intensity and its own value.

At the same time, the spatial 2-D classification branch generates character segmentation $G \in \mathbb{R}^{h \times w \times C}$ with a size of $h \times w$ and C categories. G is generated directly generated from the visual features extracted from the InceptionV3 backbone. The category is normalized using the softmax function to generate the character segmentation map.

The spatial 2-D geometry branch first uses the sigmoid activation function to generate a character localization map $Q \in \mathbb{R}^{h \times w}$ based on the same visual features as the class branch. Meanwhile, a top–down pyramid structure is employed, in which the features in the upsampling path are added to the features with the same resolution in the downsampling path to generate a sequential segmentation map. The feature map at the top of the downsampling path is encoded by the recurrent neural network (RNN) for context modeling. Following the upsampling method, two convolutional layers are used to generate the sequential segmentation map $S \in \mathbb{R}^{h \times w \times K}$, where K is the maximum sequence length. The sequential segmentation map is also normalized by the softmax function, and the sequential map $H \in \mathbb{R}^{h \times w}$ is calculated by elementwise multiplication of the kth channel of the sequential segmentation and the character localization map Q, where $H = Q \odot S^k$.

B. Sequences 1-D Feature Extraction

Temperature and spectral features exhibit a 1-D sequential distribution, where the overall sequence variation significantly affects their effectiveness. How to obtain the global and local information of sequence-based 1-D features and integrate them with spatial 2-D features is crucial for subsequent target characterization. Such information not only supplements the quality and richness of spatial features but also facilitates more accurate feature alignment. Moreover, within the cross-attention module, the 1-D sequence features can effectively guide the 2-D spatial features, further enhancing the fusion process.

Therefore, this article obtains the global and local features of the 1-D features of the sequence through the attention mechanism based on the Transformer structure, and realizes the subsequent correspondence between the 1-D features of the sequence and the 2-D features of the space through a clearer position relationship. Considering that the temperature and spectral features are long in length, for the features obtained, the 1-D feature information description of the sequence after the segmentation of the 2-D continuous frames in the space is combined. The [fir] and [last] embedding vectors are inserted at the beginning and end to mark the beginning and end of the 1-D feature information description sentence of the sequence. To ensure computational efficiency, the maximum 1-D description sequence length is set to Num. In order to learn the relative position relationship of temperature/spectral tokens in a sentence, the position vector $P \in \mathbb{R}^{\text{Num} \times D}$ is embedded into the sequence. The final sequence input to the sequence 1-D Transformer can be expressed as

$$F_\text{onedim}_t^{i,(0)} = [E(T_{\text{fir}}); E(T_1); E(T_2); \ldots; E(T_{\text{last}})] + P \tag{3}$$

where $F_\text{onedim}_t^{i,(0)}$ represents the input embedding vector of the 1-D sequence, and E represents the projection layer that maps the word vector to a D-dimensional vector. The 1-D feature embedding vector output by the final layer of the network can be expressed as $F_\text{onedim}_t^{i,(L)} = \{F_\text{onedim}_t^{\text{fir}}, F_\text{onedim}_t^{1}, F_\text{onedim}_t^{2}, \ldots, F_\text{onedim}_t^{\text{last}}\}$. The embedding vector of the last token $F_\text{onedim}_t^{\text{last}}$ will be used as the global feature F_onedim_t, which describes the 1-D sequence feature and is used to perform cross-modal matching with the 2-D spatial feature F_img_t to minimize the distance between the two.

After the 1-D features of the sequence are obtained through the Transformer, a soft attention mechanism with real-time updates is adopted. The $(K_{\text{onedim}}, Q_{\text{onedim}}, V_{\text{onedim}})$ of the 1-D feature attention mechanism of the sequence is expressed as

$$\begin{aligned} K_{\text{onedim}} &= F_\text{onedim}_t^i W^K, \quad Q_{\text{onedim}} = F_\text{onedim}_t^i W^Q \\ V_{\text{onedim}} &= F_\text{onedim}_t^i W^V. \end{aligned} \tag{4}$$

Then, the encoder output $\text{Atten}_\text{onedim}$ of the Transformer network is as follows:

$$\begin{aligned} \text{Atten}_\text{onedim}\left(y_t^i\right) &= F_\text{onedim}_t^i + \text{FFN}\left(y_t^i\right) \\ &\quad + \frac{\exp(\text{similar}_i)}{\sum_{j=1}^{K} \exp\left(\text{similar}_j\right)} \cdot V_{\text{onedim}} \end{aligned} \tag{5}$$

$$\text{FFN}\left(y_t^i\right) = \max\left(0, y_t^i W_1 + b_1\right) W_2 + b_2 \tag{6}$$

where FFN() is a feedforward neural network.

This article will obtain the encoded temperature features and spectral features as the input of the sequence 1-D features of the cross-attention mechanism module. The output of the sequence 1-D features (temperature features and spectral features) through the Transformer network is

$$\text{similar} = \cos(\theta) = \frac{Q_{\text{onedim}} \cdot K_{\text{onedim}}}{\|Q_{\text{onedim}}\| \cdot \|K_{\text{onedim}}\|} \tag{7}$$

$$F_\text{onedim} = \sum_{i=1}^{K} \left(\frac{\exp(\text{similar}_i)}{\sum_{j=1}^{K} \exp\left(\text{similar}_j\right)} \cdot V_{\text{onedim}} \right). \tag{8}$$

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Among them, the V_{onedim} of the first layer of the Transformer network selects the simulated 1-D feature to replace the V_{onedim} of the self-attention mechanism in the network.

C. Secondary Attention Module

The 2-D features are obtained through the InceptionV3 network with an added attention mechanism, and the 1-D features are obtained through the Transformer network. Different modalities exhibit modality-specific characteristics and heterogeneous distributions, while intrinsic coupling exists across modalities, making direct fusion challenging. To address this issue, the 1-D sequence features are projected into a positional space and integrated through a secondary attention mechanism, in which modality-specific importance and cross-attention jointly enable importance-aligned fusion with the spatial modality.

To enable effective interaction between 1-D semantic embeddings and 2-D spatial features, the 1-D representation is first linearly projected to match the channel dimension of the 2-D feature map and then spatially broadcast across all positions. This projection–broadcast operation provides each spatial token with access to the same semantic descriptor, forming a unified feature space in which cross-attention can establish consistent alignment between global semantics and fine-grained spatial cues with minimal computational overhead.

Since subsequent infrared small-target recognition and tracking rely primarily on spatial 2-D sequences, the cross-attention module guides the enhancement of 2-D spatial features x using the 1-D sequence features y' (temperature and spectral cues). The input to the cross-attention neural network Input_cro consists of the spatial 2-D features Atten_$img(x)$ and the 1-D sequence features Atten_$onedim(y')$

$$\begin{aligned}\text{Input_cro} &= \left\{\text{Atten_img}\,(x),\ \text{Atten_onedim}\left(y'\right)\right\}\\ &= \left\{x_1, x_2, \ldots, x_n, y'_1, y'_2, \ldots, y'_m\right\}.\end{aligned} \tag{9}$$

Then, the (K, Q, V) of the cross-attention mechanism is expressed as

$$\begin{aligned}K_{\text{Input_cro}} &= \text{Input_cro}\cdot \text{W}^K = \left(K_x, K_{y'}\right)\\ Q_{\text{Input_cro}} &= \text{Input_cro}\cdot \text{W}^Q = \left(Q_x, Q_{y'}\right)\\ V_{\text{Input_cro}} &= \text{Input_cro}\cdot \text{W}^V = \left(V_x, V_{y'}\right).\end{aligned} \tag{10}$$

Then, the cross-attention Atten_cro is as

$$\text{Atten_cro} = \text{softmax}\left(\frac{Q_{\text{Input_cro}}\ {K_{\text{Input_cro}}}^{\text{T}}}{\sqrt{d}}\right) V_{\text{Input_cro}}. \tag{11}$$

At the same time, considering that reliance on the cross-attention mechanism often overlooks the less important parts, and due to the complexity of the deep-space environment and the background, the target features may not necessarily be the most prominent. Therefore, the secondary attention mechanism is introduced to further enhance the weights and achieve a more effective alignment between the 1-D features of the sequence and the 2-D features of the space. The first weighted attention focuses on the mutual influence and alignment between features, while the second attention places more emphasis on the individual characteristics of each feature. By combining overall characteristics with their unique properties, a more thorough utilization of different modal features can be achieved in feature fusion.

The 1-D features of the sequence are sorted according to their importance, and the sorted features and the corresponding indexes of the feature positions are as follows:

$$\text{weight_onedim}\,(\text{oi}) = \text{softmax}\left(\frac{Q_{y'}{K_{y'}}^{\text{T}}}{\sqrt{d}}\right) \tag{12}$$

$$\text{Im_onedim}\,(\text{oi}) = \frac{1}{\text{To}}\sum_{\text{oi}} \text{weight_onedim}\,(\text{oi}) \tag{13}$$

$$\text{Final_onedim} = \frac{\exp\left(\text{Im_onedim}\,(\text{oi})\right)}{\sum \exp\left(\text{Im_onedim}\,(\text{oi})\right)}\cdot V_{y'} \tag{14}$$

where weight_onedim(oi) is the attention weight between the (oi)th position of the 1-D feature of the sequence. To is the number of all attention scores associated with the 1-D feature of the (oi)th sequence.

Then, the weight ranking after the spatial 2-D features is sorted as follows:

$$\text{weight_img}\,(ti, tj) = \text{softmax}\left(\frac{Q_x{K_x}^{\text{T}}}{\sqrt{d}}\right) \tag{15}$$

$$\text{Im_img}\,(ti, tj) = \frac{1}{Ti}\sum_{tj} \text{weight_img}\,(ti, tj). \tag{16}$$

The averaging in (13) and (16) is applied after softmax normalization and functions as a stabilizer against isolated noisy responses, preserving the relative contrast among salient regions. The averaging operation can stabilize the importance estimation across different positions, which improves robustness against isolated noisy activations while avoiding the introduction of additional tunable parameters

$$\text{Final_img} = \frac{\exp\left(\text{Im_img}\,(ti, tj)\right)}{\sum \exp\left(\text{Im_img}\,(ti, tj)\right)}\cdot V_x \tag{17}$$

where weight_img(ti, tj) is the attention weight of the spatial 2-D feature at position $(ti,\ tj)$. Ti is the total number of attention scores associated with the spatial 2-D feature at position $(ti,\ tj)$.

Then, the final output of the secondary attention module is as follows:

$$\begin{aligned}\text{Fusion_Atten} = \alpha\cdot\left(\text{Atten_cro}\ (x)\oplus\text{Atten_cro}\left(y'\right)\right)\\ +\beta\cdot\left(\frac{\exp\left(\text{Im_img}(ti, tj)\right)}{\sum\exp\left(\text{Im_img}\,(ti, tj)\right)}\cdot V_x\right.\\ \left.\oplus\frac{\exp\left(\text{Im_onedim}\,(\text{oi})\right)}{\sum\exp\left(\text{Im_onedim}\,(\text{oi})\right)}\cdot V_{y'}\right).\end{aligned} \tag{18}$$

The first softmax in cross-attention establishes token-level correspondence but may overemphasize dominant activations. To mitigate this, a second softmax in modality-specific self-attention normalizes intramodal importance, preventing excessive amplification and preserving secondary yet informative cues, resulting in a more balanced fusion representation.

Furthermore, the semantic branch G and the geometric localization branches Q and S supervise the shared backbone

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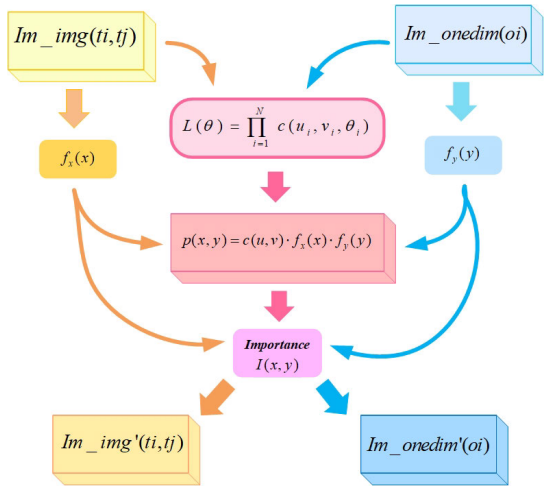


Fig. 3. Cross-modal fusion MI module. (The central part of the figure represents the updated target importance obtained based on upper tail dependence, which is used to suppress redundant features.)

from complementary viewpoints. While G highlights semantically relevant regions, Q and S contribute spatial and structural cues that exhibit stronger robustness to appearance changes. Together, these signals guide the backbone toward feature representations that are both semantically discriminative and geometrically stable, establishing reliable spatial priors for the secondary attention module. Under these priors, the second softmax balances modality-specific responses, preventing excessive attention concentration and retaining informative secondary cues.

D. MI-Driven Adaptive Modeling

MI is introduced to dynamically adjust importance weights and suppress redundant features, considering the dependence between spatial 2-D and 1-D features. The Copula method decomposes the joint distribution into marginal distributions and dependence structures, allowing separate modeling of each. The flowchart is shown in Fig. 3.

According to the features Atten_img(x) and Atten_onedim(y'), we compute their empirical cumulative distribution functions (ECDFs) $f_x(x)$ and $f_{y'}(y')$ to map the features to the interval [0, 1]

$$u = f_x(x), v = f_{y'}(y'). \tag{19}$$

In order to effectively align the spatial 2-D features and the sequence 1-D features during cross-modal fusion while preventing the impact of feature redundancy, the Gumbel Copula function is used

$$C(u,v,\theta) = \exp\left(-\left[(-\log u)^{\theta} + (-\log v)^{\theta}\right]^{\frac{1}{\theta}}\right), \quad \theta \geq 0 \tag{20}$$

where θ is the dependent parameter. When $\theta \geq 0$, the two expressions are positively correlated, and the larger the θ value, the stronger the mutual dependence. The dependence parameter θ is automatically estimated via maximum likelihood based on the observed feature distributions, allowing the intermodal dependence strength to adapt to different targets and scenes.

And then, we fit the parameters of the Copula function using maximum likelihood estimation (MLE). Then, we obtain the Copula density, as well as the marginal densities, and then calculate the joint probability density

$$p(x,y') = c(u,v)\cdot f_x(x)\cdot f_{y'}(y') \tag{21}$$

$$I(x,y) = \sum_{x^i_p}\sum_{y^i_q} p\left(x^i_p, y^i_q\right)\log\left(\frac{p\left(x^i_p, y^i_q\right)}{p\left(x^i_p\right)p\left(y^i_q\right)}\right) \tag{22}$$

where i_p and i_q represent the index of the position of the feature in the corresponding feature set.

After each calculation of the cross-attention mechanism, the importance weights are updated according to the MI between the current feature and the target. By dynamically adjusting the importance weights, the introduction of redundant information can be prevented during the feature fusion process, and the final fused features can be guaranteed to have high task relevance. Based on the MI between the 1-D feature and the 2-D feature, the 1-D feature importance and the 2-D feature importance are adjusted by weighted MI

$$\text{Im_onedim}'(\text{oi}) = \frac{\frac{1}{\text{To}}\sum_k \text{weight_onedim}(\text{oi}) \times \sum_{i_p} I(x,y')}{\sum_{i_p}\sum_{i_q} I(x,y')} \tag{23}$$

$$\text{Im_img}'(ti,tj) = \frac{\frac{1}{Ti}\sum_k \text{weight_img}(ti,tj) \times \sum_{i_q} I(x,y')}{\sum_{i_p}\sum_{i_q} I(x,y')}. \tag{24}$$

By incorporating MI, the model can achieve effective multi-feature cross-modal fusion while simultaneously adjusting the importance of modality-specific characteristics. Therefore, β is assigned a larger weight, which is further supported by the experimental results. Based on this, we set $\alpha = 0.3$ and $\beta = 0.7$ in this work. Then, the final attention Final_Atten is as follows:

$$\begin{aligned}\text{Final_Atten} = {} & \alpha\cdot\left(\text{Atten_cro}\ (x)\oplus\text{Atten_cro}\left(y'\right)\right) \\ & +\beta\cdot\left(\frac{\exp\left(\text{Im_img}'(ti,tj)\right)}{\sum\exp\left(\text{Im_img}'(ti,tj)\right)}\cdot V_x\right. \\ & \left.\oplus\frac{\exp\left(\text{Im_onedim}'(\text{oi})\right)}{\sum\exp\left(\text{Im_onedim}'(\text{oi})\right)}\cdot V_{y'}\right)\end{aligned} \tag{25}$$

where x denotes the spatial 2-D feature, y' denotes the sequence's 1-D feature in the same feature mapping space as x, (t_i, t_j) indicates the spatial position of the 2-D feature, and oi represents the (oi)th position of the 1-D feature.

We adopt an optimized triplet loss function, where positive cross-modal pairs (features of the same target across modalities) are enforced to have higher similarity than negative pairs (features of different targets), subject to a margin constraint. This formulation not only enhances the discriminability of fused representations but also explicitly suppresses redundancy by encouraging compact intraclass alignment and large interclass separability. Moreover, the margin-based design ensures robust cross-modal consistency even under noisy or degraded conditions, making it well-suited for the complex backgrounds in deep-space target recognition. The MI constraint

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is implicitly incorporated during feature fusion to regulate representation learning, while the triplet loss is applied to the resulting MI-aware features for discriminative optimization. The similarity computation and the corresponding loss function are formulated as follows:

$$\text{Simi}\left(x, y'\right) = \frac{\text{Final_img}\left(x\right) \cdot \text{Final_onedim}\left(y'\right)}{\left\|\text{Final_img}\left(x\right)\right\| \cdot \left\|\text{Final_onedim}\left(y'\right)\right\|} \tag{26}$$

$$\begin{aligned}\text{Loss} = &\max\left[0, m - \text{Simi}\left(x, y'\right) + \text{Simi}\left(\hat{x}, y'\right)\right] \\ &+ \max\left[0, m - \text{Simi}\left(x, y'\right) + \text{Simi}\left(x, \hat{y'}\right)\right]\end{aligned} \tag{27}$$

where $\hat{x}$ denotes the negative spatial 2-D feature (mismatched with y'), $\hat{y'}$ denotes the negative sequence 1-D feature (mismatched with x), and m is the boundary parameter. The dual triplet loss ensures that the similarity between matched cross-modal pairs (x, y') is higher than that of mismatched pairs $(\hat{x}, y')$ and $(x, \hat{y'})$ by at least m.

E. Infrared Target Recognition

Due to the deep-space environment, the background influencing factors are complex and discontinuous, and multiple influencing factors are coupled, which makes it highly susceptible to interference from the background environment during the recognition stage. This can lead to the target being submerged in the background or background points being mistakenly identified as the target. Therefore, the model first uses the time–space local contrast to segment the target and background areas. At the same time, in order to effectively detect the candidate target area in the image, this article adopts an automatic target frame generation method based on image preprocessing and edge detection. The candidate target frames are dynamically generated using the contour detection algorithm. In view of the characteristics of small-scale targets, the target frame size is constrained (both width and height do not exceed the specified maximum value) to improve the detection efficiency and eliminate the interference of nontarget areas.

In the target classification process, the spatial 2-D features and 1-D features are encoded into embedding vectors of the same dimension through the feature extraction module. Subsequently, a fully connected layer (dense layer) is applied to perform a nonlinear transformation on the fused features and carry out category classification

$$P = \text{softmax}\left(W \cdot \text{Final_Atten} + b\right) \tag{28}$$

where $W \in \mathbb{R}^{\text{Num}*2d}$ is the weight of the fully connected layer, Num is the number of classification categories, $2d$ is the total dimension after the spatial 2-D features and 1-D features are concatenated, $b \in \mathbb{R}^{\text{Num}}$ is the bias value, and P is the probability distribution vector of each type of target.

To achieve the priority sorting of multitarget detection, this article calculates the predicted probability of each candidate target box based on the classification network. By sorting the classification probabilities P of all candidate boxes in descending order, the top N candidate boxes with the highest probabilities (where $N \leq 8$) are selected. This strategy ensures that when processing a limited number of targets, computing resources can be concentrated on the most potentially significant candidate target boxes.

IV. Experiment and Result Analysis

A. Experimental Datasets

The four groups of datasets used in the experiment are constructed from a combination of semiphysical measurements and physics-based simulations. The semiphysical part, corresponding to Groups 2 and 3, integrates vacuum-chamber observations acquired under replicated deep-space radiance and thermal conditions with star-field backgrounds generated from the smithsonian astrophysical observatory (SAO) catalog, providing reliable temporal radiance profiles for targets with emissivities between 0.3 and 0.9. The physics-based simulation part, corresponding to Groups 1 and 4, is generated using a radiometric model that incorporates the target geometry (length: $a = 2$ m, radius: $b = 2$ m, and height: $c = 1.5$ m), inertia-driven spin motion (for example, $\omega_s = 5\pi$), and a time-varying projection-area formulation with a base area of $a_i = 0.12\pi$ and a normal vector of [0.6135, 0.3519, 0.7068]. Based on this model, Group 1 is generated under a lateral observation path, whereas Group 4 is derived from a forward observation path with spiral-like motion. Grayscale sequences are obtained through Planck-law integration, with temperature evolution governed by solar-flux absorption and emission coefficients.

1) Group 1 is the simulation data based on the characteristics of the mid-section space target. The image background is a nonuniform background without sunlight, and the target pixels are 3*3. The initial signal contrast ratio (SCR) of the images is 3, the initial signal-to-noise ratio (SNR) of the images is 2, and the data size is 256×256. The target brightness changes according to the change in the simulated radiation intensity. The initial speed of the target is 2 pixel/s, and the acceleration is 0.5 pixel/s^2. The space target direction is a horizontal straight line movement, and the interference target group movement is an irregular curve movement.
2) Groups 2 and 3 consist of semiphysical simulation data, obtained by fusing simulated images generated from the SAO star catalog with vacuum-chamber measurements. The simulated part is a space-based optical star map dataset containing moving dim and small targets, with background stars generated according to the SAO star catalog. The measured part is provided by a national-level professional laboratory, where the vacuum chamber effectively reproduces deep-space environmental conditions, ambient temperature, target distribution, and temperature variations. The reliability of these measurements is further confirmed by the close agreement between derived radiance values and real observational data. By integrating the two sources, semiphysical simulation data with realistic deep-space target characteristics are obtained. In addition, spectral and temperature variation characteristics of the space targets are derived from the vacuum-chamber experiments as sequence-level 1-D features, providing sufficient data support for this study. The space targets are aluminum alloys with emissivities of 0.3, 0.6, and 0.9. The image initial SCR is set to 3, the initial SNR to 2, and the spatial resolution to 256×256. The dataset includes 2000 training frames

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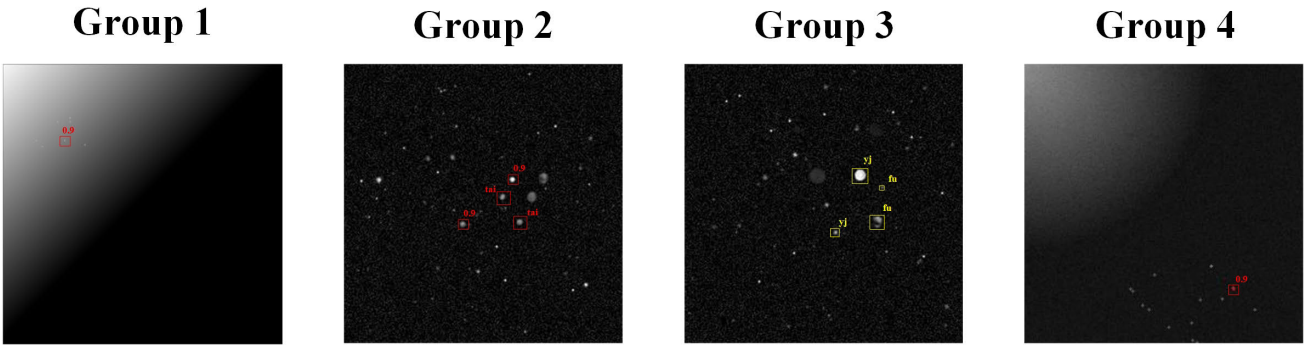


Fig. 4. Target ground truth of the four groups of images. (Four data groups with annotated target regions and material labels.)

and 800 test frames. To further increase the difficulty of recognition, the test data in Group 2 include titanium alloy (Ti alloy) targets and aluminum alloy targets coated with 0.9 emissivity paint; the emissivities of the two are close.

3) Group 4 is simulation data. The datasets are simulated by the deep-space background and imaging characteristics, simulating the flight of space targets under multiple background interference factors, where the target size is 4–5 pixels. The initial SCR of the images is 3, the initial SNR of the images is 2, and the data size is 256×256. The target brightness changes according to the change in the simulated radiation intensity. The initial speed of the target is 2 pixel/s, the target is flying in a circle, and the movement is similar to a circle. There are interferences with similar emissivity around the target, and the bait is moving irregularly.

Among them, the targets in Groups 1 and 4 are aluminum alloy objects with an emissivity of 0.9, denoted as "0.9." The targets in Group 2 include aluminum alloy with 0.9 emissivity and Ti alloy, denoted as "0.9" and "tai," respectively. Group 3 contains interference samples, with materials including acrylic glass [polymethyl methacrylate (PMMA), referred to as "yj"] and polytetrafluoroethylene (PTFE, referred to as "fu"). The ground truth of the targets in all four groups is illustrated in Fig. 4.

B. Experimental Verification Indicators

For overall classification performance, comparative experiments were evaluated with $F1$ score, Acc, area under the curve (AUC), precision, and recall. It is worth noting that in spatial target recognition, the complexity of influencing factors often leads to class imbalance. Given this challenge, AUC, being more robust to imbalance, better reflects the model's ability to detect anomalous samples and its generalization in practice. Therefore, receiver operating characteristic (ROC) curves and AUC are employed to assess whether the fused features preserve cross-modal consistency.

To evaluate the quality of feature fusion, the performance of the fused features was quantitatively evaluated using three metrics: modality complementarity gain (MCG), redundancy score (RS), and entropy. Results demonstrate that the features extracted by SA-CMF exhibit higher complementarity and lower redundancy compared to those derived from single-modality representations. Their formulations are given as follows:

$$\mathrm{MCG} = \frac{1}{M}\sum_{m=1}^{M}\left[\mathrm{sim}\left(F_f, F_m\right) - \mathrm{sim}\left(F_m, F_m\right)\right] \tag{29}$$

where F_m denotes the feature vector from modality m, F_f is the fused feature, $\mathrm{sim}(\cdot,\cdot)$ represents the cosine similarity, and M is the number of modalities. A higher MCG reflects greater cross-modal complementarity

$$\mathrm{RS} = \frac{1}{M}\sum_{m=1}^{M} I\left(F_f; F_m\right) \tag{30}$$

where $I(\cdot;\cdot)$ denotes the MI between the fused features and each individual modality. Lower RS values indicate reduced feature redundancy

$$\mathrm{Entropy}\left(F_f\right) = -\sum_{i=1}^{N} p_i \log p_i \tag{31}$$

where p_i denotes the normalized activation probability of the ith feature component in F_f, and N is the dimensionality of the fused feature. Entropy measures the information richness of the representation.

Considering engineering deployment requirements, the size and computational efficiency of the model are crucial for scientific applications such as deep-space target recognition. Therefore, in this study, we introduce four key metrics—floating-point operations (FLOPs) (to verify the theoretical computational complexity of the model), inference time (the average time required to process a single sample), computational cost (to evaluate the throughput capability of the model), and the number of parameters (the total count of learnable parameters)—to comprehensively assess the computational complexity and engineering feasibility of the proposed model.

To enable a direct and comprehensive comparison with state-of-the-art (SOTA) methods, the comparison experiment used $F1$, Acc, ROC curve, AUC value, R@1, R@5, and R@10 as experimental verification indicators. $F1$ score is particularly effective in assessing the model's capability when dealing with imbalanced class distributions, whereas Acc is more suitable for tasks with balanced class distributions. And the recall at K (R@K) metric is critical for evaluating retrieval effectiveness in practical applications. For demanding scientific tasks such as deep-space target detection, R@1, R@5, and R@10 provide

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TABLE I
PERFORMANCE COMPARISON ACROSS DIFFERENT METHODS

Data	Method	F1	Acc	AUC	Precision	Recall
Group 1	Method(1)	0.686	0.716	0.69	0.765	0.7318
	Method(2)	0.723	0.8388	0.78	0.826	0.8466
	SA-CMF	**0.8536**	**0.9233**	**0.91**	**0.861**	**0.878**
Group 2	Method(1)	0.7352	0.87	0.58	0.7142	0.6856
	Method(2)	0.802	0.9014	0.83	0.741	0.7233
	SA-CMF	**0.8823**	**0.947**	**0.93**	**0.8028**	**0.7605**
Group 3	Method(1)	0.765	0.71	0.776	0.717	0.761
	Method(2)	0.836	0.866	0.84	0.794	0.796
	SA-CMF	**0.8799**	**0.921**	**0.9**	**0.832**	**0.8188**
Group 4	Method(1)	0.8123	0.867	0.828	0.79	0.797
	Method(2)	0.865	0.901	0.86	0.867	0.852
	SA-CMF	**0.898**	**0.942**	**0.93**	**0.8971**	**0.8736**

essential benchmarks of model performance across varying detection ranges.

To further assess the robustness and reliability of the proposed model under realistic deep-space imaging conditions, three additional evaluation indicators are introduced.

The first is the performance–degradation curve (AUC). This metric evaluates the stability of recognition performance under progressively increasing image degradations such as noise, blur, or contrast loss. The integral of the degradation curve, AUC, reflects the model's global robustness to imaging quality fluctuations

$$\mathrm{AUC} = \int_0^L \mathrm{Acc}\,(l)\; dl \tag{32}$$

where l denotes the degradation level, and $\mathrm{Acc}(l)$ is the recognition Acc at level l.

Considering that sensor failures and incomplete modalities frequently occur during deep-space sensing, Robustness@Missing-Modality (R@MM) measures the retained performance when a specific modality is absent. For notational simplicity, we denote 2-D as spatial 2-D and 1-D as sequence 1-D. We report both R@MM 2-D and R@MM 1-D to evaluate the impact of missing 2-D or 1-D information

$$\mathrm{R@MM}_m = \frac{\mathrm{Acc}_{\mathrm{missing}(m)}}{\mathrm{Acc}_{\mathrm{clean}}}, \quad m \in \{\text{2-D}, \text{1-D}\} \tag{33}$$

where $\mathrm{Acc}_{\mathrm{missing}}(m)$ (Acc under modality m missing) and $\mathrm{Acc}_{\mathrm{clean}}$ (baseline Acc with clean inputs) are the key components.

To examine the reliability of prediction confidence, expected calibration error (ECE) quantifies the gap between a model's predicted confidence and its actual correctness. A lower ECE indicates better calibration, which is crucial for trustworthy decision-making in scientific applications

$$\mathrm{ECE} = \sum_{k=1}^{K} \frac{|B_k|}{n}\, |\mathrm{Acc}\,(B_k) - \mathrm{Conf}\,(B_k)| \tag{34}$$

where B_k denotes the kth confidence bin, and n is the total number of samples.

And to evaluate the influence of varying N on the selection of candidate boxes, we introduce three widely used spatial-quality indicators. Best intersection over union (IoU) quantifies the maximal overlap between the selected candidates and the ground-truth box, reflecting the upper bound of localization Acc under a given N. *Recall* measures the proportion of ground-truth targets that are successfully preserved within the top-N candidates, indicating whether essential detection information is retained. *Coverage* assesses how comprehensively the selected candidates span all true targets, capturing the completeness and robustness of candidate selection in multiobject scenarios.

C. Experimental Validation of Model Components (Ablation Study)

1) Ablation Study of Model Components: To assess the contribution of each module to feature alignment and fusion, we conducted ablation studies on four model variants: 1) baseline with cross-attention only (Method 1); 2) cross-attention with single-modality importance weighting (Method 2); and 3) the full model with MI integration. Results on the validation dataset highlight the role of the secondary attention module and the MI. The results are shown in Fig. 5 and Table I.

The cross-attention-only model fails to capture modality-specific features, leading to misidentification among targets with similar emissivity. Adding single-modality weighting alleviates this issue but still suffers from redundancy without MI. In contrast, the complete model fully leverages modality characteristics and their interactions through secondary attention and dynamic adjustment, achieving accurate target–bait discrimination. MI enables adaptive weighting of modalities, reducing redundancy and ensuring effective feature fusion. Compared to other groups, although the results of Group 2 show slightly lower precision and recall due to class imbalance and complex backgrounds, the reduction is minor. Overall, Table I demonstrates that our model maintains stable and superior performance across diverse datasets.

By comparing four groups with varying SNRs and spectral feature quality, we verify that the proposed model consistently achieves strong performance under different data conditions, enabling accurate recognition of space targets. Among the groups, Group 2 best illustrates the challenges of complex

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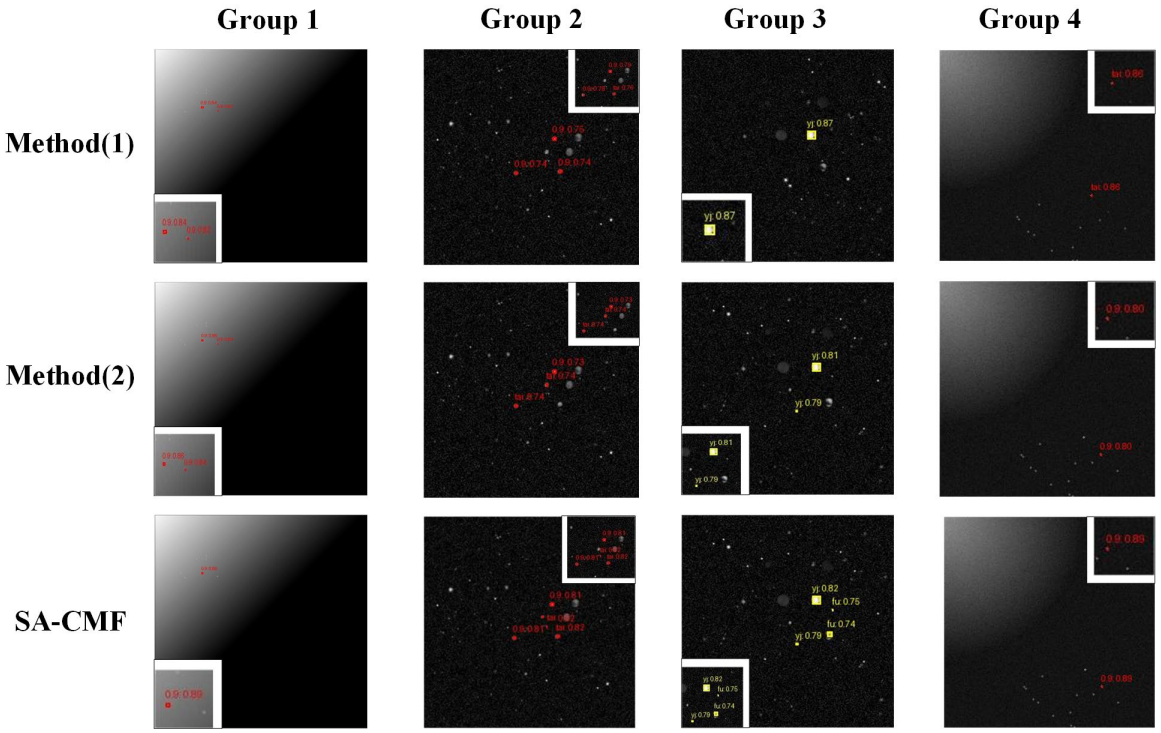


Fig. 5. Model ablation experimental results on four groups.

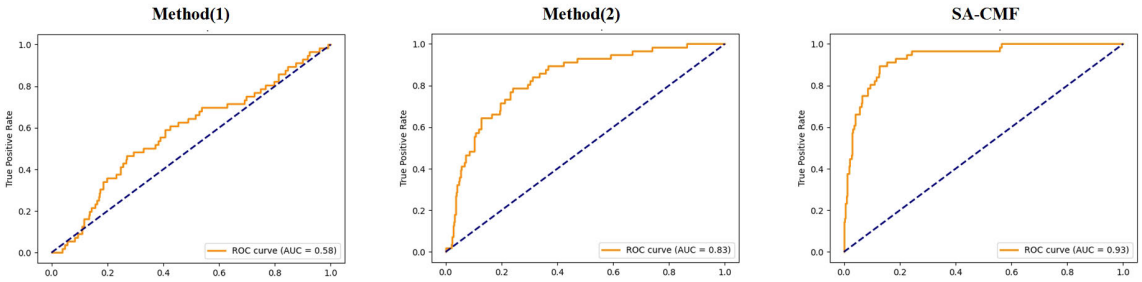


Fig. 6. Model ablation experimental results on Group 2.

deep-space backgrounds and interfering objects, and its results (Fig. 6) are mainly used for performance demonstration.

As evidenced by Fig. 6, the ROC curves and corresponding AUC values indicate superior generalization capability, even under challenging conditions such as complex backgrounds and low SNR in images, confirming its robust discriminative power.

2) Model Sensitivity to Parameters: Different parameter combinations also have a significant impact on model performance. In (25), the weights α and β represent the relative contributions of modality-specific features and intermodality relationships to the fused representation, respectively. To enhance adaptability, we introduce MI into the design, enabling the model to dynamically adjust the importance of different modalities. This mechanism ensures that modality-specific features are fully exploited while adaptively regulating their influence on the fused features. To evaluate this effect, we conducted ablation studies by varying the parameter settings of α and β as PS 1:(0.7/0.3), PS 2:(0.5/0.5), and PS 3:(0.3/0.7). The result is shown in Table II.

As shown in Table II, the best performance is achieved when the weights α and β are set to 0.3 and 0.7, respectively. This result validates the effectiveness of incorporating MI and highlights the importance of adaptively adjusting the relative influence of different modality-specific features on the fused representation. Such an adaptive weighting strategy plays a critical role in enhancing the overall performance of the proposed model.

3) Computational Efficiency and Model Complexity Analysis: In deep-space applications, multiplatform collaboration requires models to balance parameter size, inference efficiency, and computational cost. These factors determine whether a cross-modal fusion model can achieve effective feature integration while remaining deployable. To this end, we evaluate the proposed model in terms of FLOPs, inference time, computational cost, and parameter count. The experimental results are summarized in Table III.

As shown in Table III, compared with Method 2 (9.97G, 6.00 ms), SA-CMF introduces only a slight increase in FLOPs (11.45G) and inference latency (12.3 ms). Nevertheless, it achieves the highest throughput (166.79 samples/s) and a

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TABLE II
PERFORMANCE COMPARISON UNDER DIFFERENT PARAMETER SETTINGS

Parameters Setting	MCG	Redundancy Score			Entropy		
		1D-2D RS	2D-fusion RS	1D-fusion RS	2D Entropy	1D Entropy	Fusion Entropy
PS 1	0.3009	0.4621	0.1561	0.1416	9.1621	9.7300	2.0167
PS 2	0.4328	0.4621	0.0866	0.0511	5.6710	5.9121	1.3067
PS 3	**0.5598**	**0.4621**	**0.0251**	**0.0176**	**5.0133**	**5.7367**	**1.2957**

TABLE III
COMPARISON OF FLOPS, INFERENCE TIME, COMPUTATIONAL COST, AND PARAMETERS AMONG DIFFERENT METHODS

Method	FLOPs		Inference Time (ms)	Computational Cost (samples/s)	Params (M)	
	Fusion (M)	Total (G)			Fusion	Total
Method (1)	—	5.6	3.93	96.18	—	19.26
Method (2)	4.21	9.97	6.00	128.75	4.21	23.87
SA-CMF	**4.21**	**11.45**	**12.3**	**166.79**	**4.21**	**47.74**

substantially larger representational capacity (47.74M parameters), indicating a well-balanced tradeoff between efficiency and Acc. Notably, the fusion module itself accounts for only a negligible fraction of the overall FLOPs, implying that the additional computational cost mainly arises from enhanced feature representation rather than fusion overhead. Overall, SA-CMF demonstrates high computational efficiency and is well-suited for engineering applications requiring both Acc and high processing speed.

With secondary attention and MI, parameters rise from 19.26M to 47.74M due to adaptive cross-modal weighting. However, Table III shows only a slight increase in inference time, indicating that the added complexity enhances fusion without sacrificing real-time performance.

4) Sensitivity Analysis of Identification Parameters: For the classification Acc, stability, and reliability of the detection results, the choice of the candidate-box upper bound N is equally critical. To validate the threshold adopted in this article, we conduct an experimental analysis by varying $N \in [1, 20]$ and examining the corresponding trends of *Best IoU*, *Recall*, and *Coverage*. This evaluation allows us to assess how the spatial-quality indicators change with N, thereby verifying the rationality and robustness of the selected threshold. The results are shown in Fig 7.

Based on the experimental results, all three metrics exhibit a clear stabilization trend once the number of candidate boxes exceeds $N > 8$, with no further significant performance gains. And when $N = 8$, the values of *Best IoU*, *Recall*, and *Coverage* are already close to their respective plateaus, indicating that this setting provides a favorable balance between recognition Acc and computational efficiency.

D. Model Robustness and Generalization Ability Verification

1) Robustness Evaluation Under Hardware-Induced Degradations: In deep-space exploration, hardware limitations often introduce challenges such as inconsistent sampling frequencies, low SNR with grayscale instability, and incomplete modalities. These issues lead to degraded data quality and undermine cross-modal consistency.

TABLE IV
ABLATION STUDY RESULTS UNDER DIFFERENT DEGRADATION CONDITIONS

Method	MCG	RS	Entropy	F1	Acc
DDSF	0.4319	0.5962	7.0377	0.7632	0.7988
LSGI	0.4207	0.5433	7.1981	0.7514	0.7832
MI(1)	0.5037	0	8.0726	0.5647	0.5964
MI(2)	0.4866	0	7.6549	0.5203	0.5733
SA-CMF	**0.5598**	**0.4621**	**1.2957**	**0.8823**	**0.947**

To evaluate the feasibility and robustness of SA-CMF under hardware-related constraints, we simulate three degradation types: degraded data with sparse frames (DDSF) (missing or irregular frames), low signal-to-noise ratio imaging (LSGI) (low-SNR imaging via Gaussian and salt-and-pepper noise), and MI [removal of 2-D or 1-D features, denoted MI(1) and MI(2)]. The results are summarized in Table IV.

The results show that SA-CMF remains robust under severe hardware-related degradations. In DDSF and LSGI settings, the model maintains low redundancy and entropy, indicating stable feature extraction despite missing frames and extremely low SNR. The $F1$ and Acc scores also demonstrate strong recognition performance under these challenging conditions.

For MI(1) and MI(2), although removing one modality reduces $F1$ and Acc, the model still preserves effective feature usage and controlled information entropy. This confirms that SA-CMF retains practical reliability even when facing incomplete sensing or other deep-space hardware limitations.

2) Ablation Analysis of Softmax Mechanism: To evaluate whether the second softmax enhances cross-modal comple-

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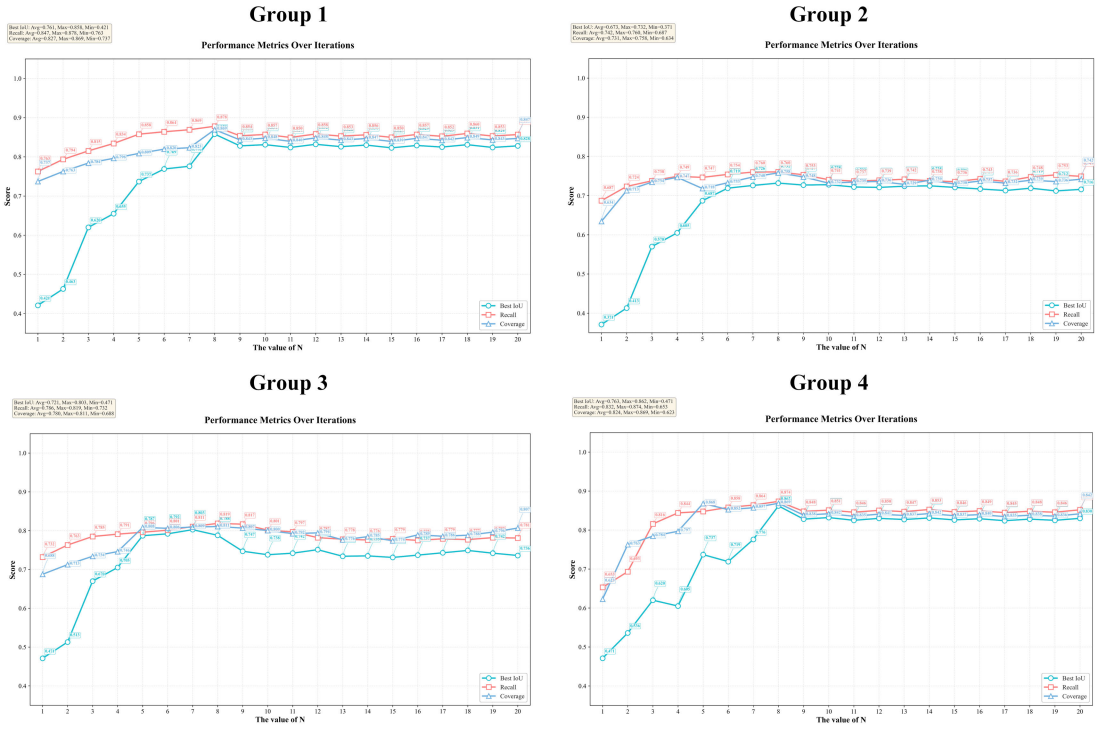


Fig. 7. Trends of *Best IoU*, *Recall*, and *Coverage* with varying candidate-box upper bound N across the four test sets. These results demonstrate how the detection performance evolves as N changes, thereby validating the stability and robustness of the chosen threshold.

TABLE V
ABLATION STUDY ON THE SECOND SOFTMAX MECHANISM

Method	MCG	Redundancy Score	Fusion Entropy	F1	Acc	AUC	ECE
RSS	0.3603	0.7311	4.5177	0.7545	0.7816	0.76	0.1225
SA(0.5)	0.5271	0.5381	2.6972	0.8377	0.9124	0.86	0.0736
SA(2.0)	0.5399	0.4208	1.0629	0.8568	0.9308	0.89	0.0524
AG	0.2851	0.6392	5.6393	0.8428	0.9165	0.91	0.0649
SA-CMF	**0.5598**	**0.4621**	**1.2957**	**0.8823**	**0.947**	**0.93**	**0.0464**

mentarity without disrupting the model's core behavior, we introduce three ablations: 1) remove second softmax (RSS), eliminating position-level normalization; 2) sharpness ablation (SA), applying a temperature $\tau \in \{0.5, 2.0\}$ to adjust attention sharpness; and 3) alternative gating (AG), replacing the second softmax with a sigmoid gate. These variants assess their influence on fusion balance and recognition stability.

Table V shows that without the second softmax, the model's redundancy suppression drops markedly, feature uncertainty increases, and recognition Acc declines. Adjusting the attention sharpness further confirms that the softmax's normalization is crucial for performance. Compared to other gating mechanisms, it more effectively captures salient features. This demonstrates that the second softmax, applied via modality-specific self-attention, normalizes intramodal importance while preserving secondary yet meaningful features, yielding a balanced and robust fusion.

E. Experimental Validation of Fusion Feature Effectiveness and Redundancy Suppression

To verify the effectiveness of the proposed algorithm in enhancing feature utility and suppressing redundancy, we adopt the same three model configurations used in the ablation studies: a baseline model using only cross-attention, a model incorporating both cross-attention and single-modality importance weighting, and the complete model with additional MI integration.

As shown in Table VI, the fused features obtained by the proposed SA-CMF model exhibit strong modality complementarity. By incorporating both cross-attention and single-modality importance weighting mechanisms, the model effectively reduces the entropy of the fused features, resulting in more accurate and discriminative target representations. Furthermore, the introduction of MI significantly suppresses redundancy, enabling the model to better leverage cross-modal complementary information and thereby enhance the effectiveness of the fused features.

To evaluate feature clustering and separability across categories or modalities, we applied t-distributed stochastic neighbor embedding (t-SNE) to project high-dimensional features into two dimensions for visualization, highlighting their discriminative power. To demonstrate the advantage of fused features over single-modal ones, we visualized 1-D, 2-D,

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TABLE VI
COMPARISON OF MCG, RS, AND ENTROPY AMONG DIFFERENT METHODS

Method	MCG	Redundancy Score			Entropy		
		1D-2D	2D-fusion	1D-fusion	2D	1D	Fusion
Method (1)	0.3139	0.4621	0.1637	0.1352	8.4366	9.1123	2.6612
Method (2)	0.4759	0.4621	0.1252	0.1192	6.4977	7.6169	1.4736
SA-CMF	**0.5598**	**0.4621**	**0.0251**	**0.0176**	**5.0133**	**5.7367**	**1.2957**

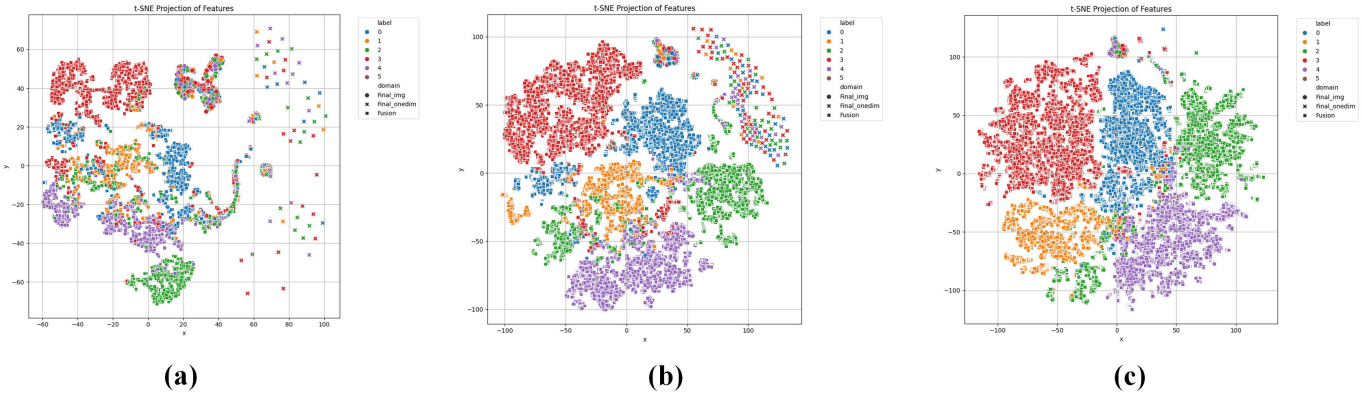


Fig. 8. t-SNE projection of features on three methods. (a) Clustering and separability of both single-modal and fused features obtained from Method 1. (b) Results based on Method 2. (c) Clustering and separability derived from the complete model proposed in this article.

and fused feature representations using t-SNE, as shown in Fig. 8.

Fig. 8 shows that incorporating the secondary attention and MI modules significantly enhances interclass separability and intraclass compactness. Compared to single-modal features, the fused features form more distinct and compact clusters, validating the effectiveness of the proposed secondary attention-based fusion mechanism for target classification.

F. Experimental Comparison of Models for Space Target Recognition

1) Comparison of Feature Extraction and Redundancy Reduction Algorithms: To assess the contribution of our feature extraction design, we compare different combinations of 2-D and 1-D extractors, as both greatly influence feature validity and the overall fusion performance. For spatial 2-D features, we adopt Swin Transformer and VT [54], [55], denoted as ST and VT, and ConvNeXt [56]. For sequence 1-D features, we employ two LLM-based extractors [57], [58] (denoted as I-LLM and LLM-T) as well as the classical method proposed by PAN et al. (denoted as PAN) [59]. These comparisons allow us to verify the effectiveness and robustness of the proposed extraction module within our fusion framework.

And our model dynamically adjusts the importance of each modality, enabling adaptive extraction of informative features while suppressing redundancy introduced during fusion. To assess the effectiveness of our redundancy-reduction strategy, we compare it with three representative approaches. CSF-Mamba [60] employs selective cross-state fusion to filter redundant multimodal cues. RSRWKV [61] focuses attention on essential spatial dependencies through a linear 2-D mechanism. Tensor-product multimodal network Tensor Product Fusion (TPF) [62] model structured cross-modal interactions and naturally limit redundant feature contributions. These methods form compact yet meaningful baselines for evaluating the advantages of our MI–Gumbel-guided fusion.

Based on the results Table VII, most 2-D feature acquisition methods struggle with small deep-space targets, as their spatial footprint is extremely limited and difficult to highlight. Likewise, spectral cues rely heavily on peak–valley structures, in which most 1-D encoders cannot fully exploit. In contrast, our Transformer-based design captures both global and local dependencies in the 1-D sequence and establishes clearer positional correspondence with 2-D spatial features. This leads to more effective feature extraction, reduced redundancy, and richer fused representations. Across all metrics, the proposed model consistently outperforms recent baseline methods, demonstrating superior feature utilization and overall recognition performance.

And the results show that, although these existing fusion methods improve feature selection, they still rely on fixed structural priors and cannot explicitly distinguish informative dependence from redundant correlation. In contrast, our MI-guided mechanism directly measures modalitywise statistical dependence and dynamically updates importance weights, enabling finer grained suppression of redundant overlap and more reliable task-relevant fusion. Moreover, the improvements in Acc and other evaluation metrics confirm that the MI-selected features contribute more reliably to the final target recognition performance.

2) Experimental Comparison of Cross-Modal Fusion Algorithms: To verify the effectiveness of the secondary attention module and the MI-based multifeature alignment and modal

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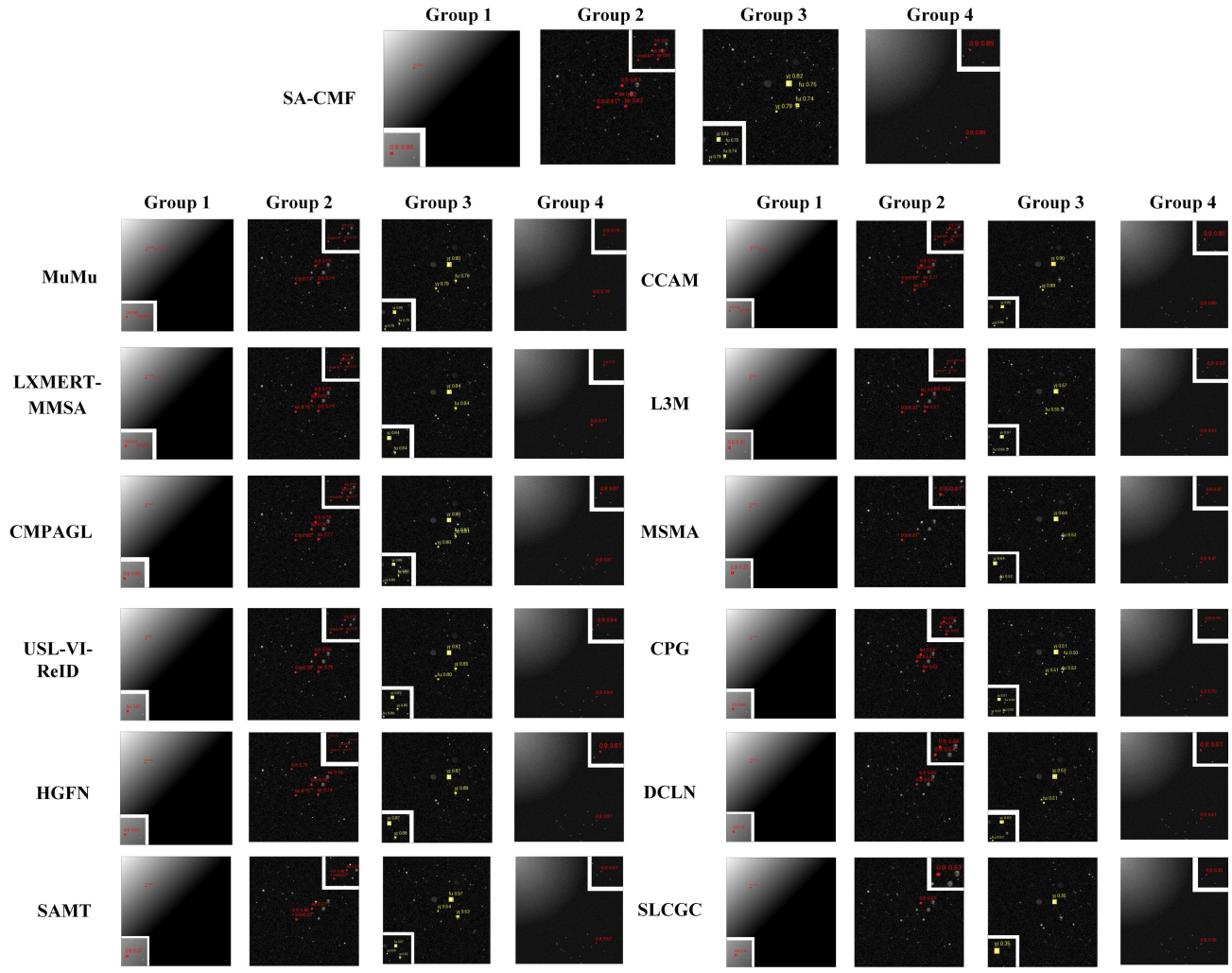


Fig. 9. Comparison of the experimental results on four groups.

TABLE VII
COMPARISON OF FEATURE EXTRACTION AND REDUNDANCY REDUCTION ALGORITHMS

Method	MCG	1D-2D RS	2D-fusion RS	1D-fusion RS	2D Entropy	1D Entropy	Fusion Entropy	F1	Acc
2D Feature Acquisition Method									
ST	0.3073	0.6377	0.2941	0.2612	10.6247	12.3611	3.5662	0.6389	0.6815
Vt	0.2706	0.7982	0.3058	0.2864	11.0382	12.8783	3.7654	0.6019	0.6960
ConvNeXt	0.4218	0.5750	0.1364	0.2318	9.6361	10.6803	2.2787	0.8192	0.8599
1D Feature Acquisition Method									
I-LLM	0.3911	0.6984	0.2866	0.2016	10.6973	11.1387	3.4631	0.6834	0.7018
LLM-T	0.4051	0.5916	0.2530	0.1347	8.7668	9.6955	2.0627	0.7762	0.8217
Pan	0.3687	0.6728	0.2901	0.1982	9.4582	10.0387	3.1541	0.7035	0.7366
Redundancy Reduction Method									
CSFMamba	0.2369	0.7365	0.3875	0.3012	12.3740	11.7630	4.0074	0.6022	0.6542
RSRWKV	0.3732	0.7528	0.3296	0.2977	12.0617	11.5441	3.8974	0.6514	0.6792
TPF	0.4377	0.5962	0.1658	0.1548	10.6332	10.2074	3.4103	0.7228	0.7051
SA-CMF	**0.5598**	**0.4621**	**0.0251**	**0.0176**	**5.0133**	**5.7367**	**1.2957**	**0.8823**	**0.9470**

fusion strategy, ten comparison algorithms were selected: MuMu [33], which effectively uses contrastive learning for multimodal feature alignment; LXMERT-MMSA [36], which aligns sequence 1-D and spatial 2-D features by mining

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TABLE VIII
PERFORMANCE, EFFICIENCY, AND COMPARATIVE ANALYSIS OF DIFFERENT METHODS ACROSS FOUR GROUPS

Method	Group 1		Group 2		Group 3		Group 4		FLOPs	Inference Time
	F1	Acc	F1	Acc	F1	Acc	F1	Acc	(G)	(ms)
SA-CMF	**0.8536**	**0.9233**	**0.8823**	**0.9470**	**0.8799**	**0.9210**	**0.8980**	**0.9420**	**11.45**	**12.3**
MuMu	0.7293	0.8124	0.8017	0.8270	0.8532	0.8749	0.8641	0.8360	34.7	123
LXMERT-MMSA	0.7170	0.8046	0.7966	0.8861	0.8014	0.8854	0.8160	0.8470	24.7	**8.9**
CMPAGL	0.8369	0.9177	0.7945	0.8470	0.8277	0.8633	0.8670	0.9130	27.86	17.3
USL-VI-ReID	0.8270	0.9061	0.8638	0.9129	0.8655	0.9017	0.8542	0.9080	30.1	23
HGFN	0.8085	0.8799	0.8342	0.8966	0.8311	0.8865	0.8271	0.8863	**2.7**	18.1
CCAM	0.7855	0.8677	0.8139	0.8742	0.8322	0.8711	0.8323	0.8398	12.7	23
L3M	0.8431	0.8556	0.8913	0.8973	0.8307	0.8273	0.8469	0.8532	27.3	19.16
MSMA	0.602	0.643	0.6551	0.6743	0.5806	0.5977	0.6071	0.6498	40.2	21.08
CPG	0.7372	0.7447	0.7774	0.7916	0.6374	0.6768	0.6355	0.6716	36.16	30.63
SAMT	0.761	0.8022	0.8594	0.8003	0.6892	0.7095	0.6693	0.6918	19.5	26.81
DCLN	0.5692	0.6217	0.6370	0.6877	0.5712	0.6233	0.6897	0.7231	52.3	69.7
SLCGC	0.4216	0.5796	0.6018	0.6742	0.5297	0.5968	0.6713	0.7108	5.6	10.5

deep semantic information; HGFN [10], which constructs a joint visual–textual graph model for effective alignment; USL-VI-ReID [38], which utilizes channel enhancement to reduce modal discrepancies; a cross-channel attention module [39], abbreviated as CCAM, which enables early alignment across different modalities; and CMPAGL [40], which performs cross-modal feature alignment by combining global and local information. In addition, we further include four recent multistage or secondary-attention-based methods: CPG [47], which leverages prototype-guided multilayer alignment; SAMT [49], which introduces a secondary-attention multistage Transformer structure; L3M [52], which employs lightweight multiscale multiattention mechanisms; and MSMA [53], which integrates multistage multiattention for multimodal feature interaction. These algorithms are used to comprehensively verify the feature alignment capability of the proposed model across four validation datasets. To verify the effectiveness of the proposed fusion module, we replace the multimodal fusion components in the proposed SA-CMF module with the DCLN [41] and SLCGC [42], and evaluate whether it can more effectively capture complementary cross-modal features for accurate recognition. The results are shown in Fig. 9.

The comparative analysis employed several metrics: $F1$ score, Acc, recall at K (R@1, R@5, and R@10). By combining $F1$ score and Acc, we can more effectively demonstrate that the fused features obtained by our model exhibit high recognition capability across different application scenarios. The comparison experimental results are shown in Tables VIII and IX.

Base on Tables VIII and IX, the proposed algorithm achieves consistently superior performance across all four datasets, particularly in distinguishing spectrally similar targets, such as high-emissivity objects ($\varepsilon = 0.9$) versus Ti alloys, and acrylic glass versus PTFE baits—cases where methods like MuMu and LXMERT-MMSA frequently misclassify due to insufficient modeling of modality-specific features. Although most approaches perform well on Group 4, our model further improves recognition confidence, demonstrating stronger discriminative capability. Compared with recent multistage or secondary-attention frameworks, our method maintains higher Acc across all settings while preserving low computational cost, which is essential for deep-space target recognition with strict resource constraints. This advantage arises from our secondary attention mechanism, which not only emphasizes modality-unique cues but also captures cross-modal dependencies to maximize feature utility, thereby enabling more stable and effective recognition. And the experimental results indicate that the proposed fusion model jointly considers intermodal relationships and modality-specific importance, leading to more effective multimodal feature extraction and more complete cross-modal feature complementarity.

To further strengthen the evaluation of our method, three robustness-oriented indicators were additionally introduced: AUC degrade, R@MM, and ECE. AUC degrade reflects the model's stability under increasing image degradations, R@MM measures its performance retention when one sensing modality is missing, R@MM 2-D measures recognition robustness when 2-D features are absent, and R@MM 1-D measures robustness when textual features are removed. ECE evaluates the reliability of its confidence calibration. Together, these metrics provide a more comprehensive assessment of the model's robustness and prediction trustworthiness. The comparison results are presented in Table X.

Through comparisons with the latest and mainstream approaches, we evaluated the model's stability, robustness under missing modalities, and calibration reliability. The results demonstrate that the proposed method not only achieves superior Acc and classification performance for space target recognition but also maintains strong stability when data quality degrades. Even when one modality is absent, the model still preserves competitive recognition capability and robustness. Furthermore, the low ECE score confirms that the predicted confidence is well-aligned with actual correctness, illustrating the reliability of the model's decision-making process.

G. Limitation and Future Works

Although the framework performs well, it still depends on the quality of features extracted by each modality. When

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TABLE IX
RECALL AT TOP-K (R@1, R@5, AND R@10) PERFORMANCE COMPARISON ACROSS FOUR GROUPS

Method	Group 1			Group 2			Group 3			Group 4		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
SA-CMF	**0.7022**	**0.7962**	**0.8834**	**0.7314**	**0.8060**	**0.8972**	**0.7104**	**0.7977**	**0.8865**	**0.7433**	**0.8012**	**0.8918**
MuMu	0.6825	0.6948	0.8012	0.7011	0.7455	0.8543	0.6511	0.7415	0.8270	0.6704	0.7543	0.8430
LXMERT-MMSA	0.6033	0.7316	0.7901	0.7017	0.7678	0.8490	0.6724	0.7288	0.8315	0.6911	0.7601	0.8511
CMPAGL	0.6830	0.7688	0.7850	0.6866	0.7467	0.8312	0.6698	0.7691	0.8458	0.7261	0.7903	0.8736
USL-VI-ReID	0.6854	0.7651	0.8409	0.7370	0.8052	0.8922	0.7020	0.7833	0.8510	0.7297	0.7977	0.8804
HGFN	0.6740	0.7022	0.8017	0.7004	0.7911	0.8547	0.6832	0.7588	0.8072	0.7099	0.7537	0.8688
CCAM	0.6524	0.7364	0.8033	0.7071	0.7647	0.8651	0.6722	0.7499	0.8038	0.6977	0.7434	0.8198
L3M	0.6893	0.7001	0.779	0.7088	0.7311	0.8295	0.6611	0.7577	0.8616	0.7218	0.7895	0.8599
MSMA	0.5361	0.5688	0.6058	0.5912	0.6388	0.6917	0.5931	0.6383	0.7347	0.6099	0.6639	0.7978
CPG	0.572	0.5963	0.629	0.639	0.6614	0.736	0.5862	0.6433	0.7688	0.6287	0.6784	0.7649
SAMT	0.6122	0.6488	0.6983	0.6927	0.7619	0.8013	0.6518	0.7394	0.8319	0.6977	0.7619	0.8449
DCLN	0.5287	0.5639	0.5714	0.5653	0.5892	0.6217	0.5428	0.5872	0.6018	0.5719	0.5967	0.6108
SLCGC	0.4768	0.4919	0.5267	0.5227	0.5649	0.5895	0.5165	0.5532	0.5774	0.5377	0.5669	0.6007

TABLE X
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON FOUR VALIDATION GROUPS

Method	Group1			Group2			Group3			Group4		
	AUC	R@MM 2D 1D	ECE	AUC	R@MM 2D 1D	ECE	AUC	R@MM 2D 1D	ECE	AUC	R@MM 2D 1D	ECE
SA-CMF	**0.91**	**0.1426** **0.1377**	**0.0397**	**0.93**	**0.1375** **0.1339**	**0.0464**	**0.9**	**0.1348** **0.1287**	**0.0639**	**0.93**	**0.1352** **0.1317**	**0.0717**
MuMu	0.77	0.0549 0.0526	0.1018	0.79	0.0688 0.0593	0.1189	0.75	0.0492 0.0475	0.1269	0.72	0.0512 0.0483	0.1277
LXMERT-MMSA	0.73	0.0458 0.0411	0.1346	0.75	0.0455 0.0467	0.1203	0.69	0.0324 0.0333	0.1695	0.68	0.0377 0.0338	0.1860
CMPAGL	0.76	0.0464 0.0426	0.1208	0.65	0.0489 0.0415	0.1139	0.69	0.0345 0.0261	0.1364	0.64	0.0358 0.0313	0.1433
USL-VI-ReID	0.63	0.0368 0.0344	0.1437	0.69	0.0351 0.0323	0.1288	0.69	0.0284 0.0313	0.1577	0.63	0.0291 0.0274	0.1597
HGFN	0.89	0.1075 0.1033	0.0882	0.87	0.1218 0.1174	0.0921	0.81	0.0767 0.0727	0.0966	0.82	0.0636 0.0586	0.1030
CCAM	0.76	0.0774 0.0747	0.0991	0.79	0.0751 0.0725	0.1167	0.77	0.0616 0.0523	0.1028	0.75	0.0604 0.0593	0.1228
L3M	0.91	0.1123 0.1073	0.0760	0.89	0.1065 0.1078	0.0899	0.82	0.0773 0.0681	0.0950	0.79	0.0611 0.0544	0.0980
MSMA	0.75	0.0496 0.0431	0.1283	0.74	0.0423 0.0425	0.1366	0.71	0.0361 0.0315	0.1479	0.71	0.0337 0.0291	0.1562
CPG	0.62	0.0322 0.0307	0.1683	0.69	0.0341 0.0291	0.1427	0.63	0.0269 0.0286	0.1680	0.65	0.0283 0.0273	0.1697
SAMT	0.66	0.0487 0.0462	0.1335	0.63	0.0427 0.0399	0.1398	0.61	0.0358 0.0334	0.1761	0.66	0.0346 0.0317	0.1730
DCLN	0.56	0.0114 0.0121	0.1876	0.58	0.0128 0.0117	0.1674	0.59	0.0177 0.0173	0.1642	0.60	0.0217 0.0224	0.1461
SLCGC	0.52	0.0107 0.0112	0.1962	0.54	0.0121 0.0124	0.2006	0.55	0.0168 0.0154	0.1889	0.56	0.0255 0.0263	0.1843

the encoders fail to obtain clear structural information, such as in heavily blurred images or very short sequence 1-D descriptions, the fusion module cannot fully compensate for the weakened inputs. In addition, the use of modality-specific attention increases computation compared with a single-attention design, which may affect use on devices with strict resource limits.

In future work, we will further enhance the framework's reliability in modality-deficient or degraded conditions. Building on our current design, we plan to incorporate a

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redundancy-aware reconstruction branch capable of estimating surrogate representations when a modality becomes unavailable, as well as an uncertainty-guided gating mechanism that dynamically adjusts fusion weights based on modality confidence. These extensions are expected to strengthen the stability of the cross-modal fusion process and improve the overall adaptability of the model across diverse operational environments.

V. Conclusion

In deep-space target recognition, the key challenge of multimodal feature fusion is to preserve modality-specific characteristics while disentangling and exploiting intermodal dependencies. To address this, we propose an SA-CMF framework. The framework employs a secondary attention structure that aligns features across modalities through a cross-attention module, while assigning individual attention to capture modality-specific importance. MI is further introduced to dynamically refine the fusion process, suppressing redundancy and improving adaptability to complex environments. Given the highly variable nature of deep-space observations, this design helps the model maintain stable cross-modal interactions: the secondary attention mechanism strengthens modality-unique cues, while MI-driven weighting highlights informative features and reduces the influence of unreliable ones. Experiments show that SA-CMF achieves 94.7% recognition Acc and maintains stable performance across diverse datasets, indicating its potential for reliable multimodal sensing in deep-space scientific exploration.

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